



Customer Churn Analytics for the Banking Industry

SQL-Driven Customer Retention Insights Using PostgreSQL

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Understanding the Customer Churn Problem

- ❖ Customer churn is one of the biggest challenges in the banking industry. Losing existing customers impacts profitability, customer lifetime value, and long-term business growth.
- ❖ This analysis aims to identify customer segments and behavioral patterns associated with churn using SQL-based exploration in PostgreSQL.





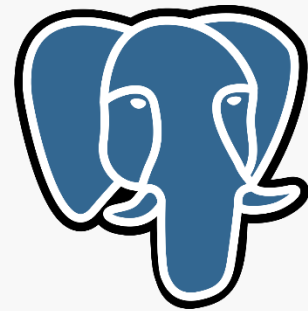
Dataset Overview & Customer Attributes

The screenshot shows a PostgreSQL query editor interface. The query entered is `SELECT * FROM churn LIMIT 10;`. The results are displayed in a table with 10 rows and 11 columns. The columns are: rownumber, customerid, surname, creditscore, geography, gender, age, tenure, balance, and numofproducts. The data shows various customer attributes such as age, gender, and balance.

rownumber	customerid	surname	creditscore	geography	gender	age	tenure	balance	numofproducts
1	15634602	Hargrave	619	France	Female	42	2	0	1
2	15647311	Hill	608	Spain	Female	41	1	83807.86	1
3	15619304	Onio	502	France	Female	42	8	159660.8	3
4	15701354	Boni	699	France	Female	39	1	0	2
5	15737888	Mitchell	850	Spain	Female	43	2	125510.82	1
6	15574012	Chu	645	Spain	Male	44	8	113755.78	2
7	15592531	Bartlett	822	France	Male	50	7	0	2
8	15656148	Obinna	376	Germany	Female	29	4	115046.74	4

Category	Attributes
Demographic	Geography, Gender, Age
Financial	Credit Score, Balance, Salary
Behavioral	Active Member, Products
Target Variable	Exited

*The dataset combines demographic, financial, and behavioral indicators to evaluate customer churn patterns.



PostgreSQL Data Engineering & Query Execution

bank_churn/postgres@PostgreSQL 18

Query

```
1 SELECT * FROM churn LIMIT 10;
2
3 SELECT exited, COUNT(*)
4 FROM churn
5 GROUP BY exited;
6
7 SELECT AVG(creditscore)
8 FROM churn;
9
10 SELECT AVG(balance)
11 FROM churn;
12
13 SELECT geography, COUNT(*)
14 FROM churn
15 WHERE exited = 1;
```

Data Output

	rownumber integer	customerid bigint	surname character varying (100)	creditscore integer	geography character varying (50)	gender character varying (20)	age integer	tenure integer	balance numeric	numofproducts integer
1	1	15634602	Hargrave	619	France	Female	42	2	0	1
2	2	15647311	Hill	608	Spain	Female	41	1	83807.86	1
3	3	15619304	Onio	502	France	Female	42	8	159660.8	3
4	4	15701354	Boni	699	France	Female	39	1	0	2
5	5	15737888	Mitchell	850	Spain	Female	43	2	125510.82	1
6	6	15574012	Chu	645	Spain	Male	44	8	113755.78	2
7	7	15592531	Bartlett	822	France	Male	50	7	0	2
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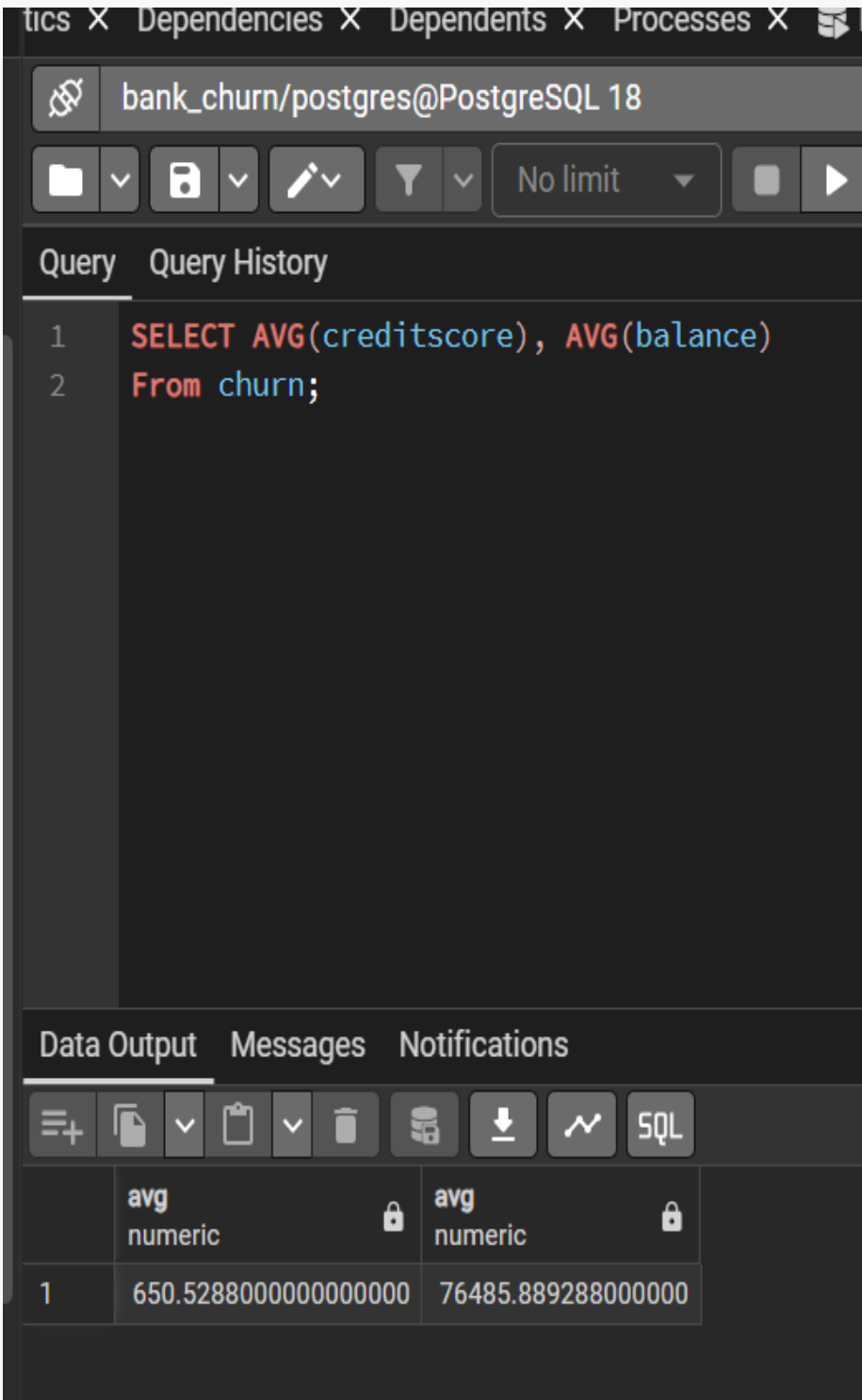
Total rows: 10 Query complete 00:00:00.126 CRLF Ln 1, Col 30

SQL Operations used :

- ❖ CREATE TABLE
- ❖ COPY
- ❖ SELECT
- ❖ GROUP BY
- ❖ Aggregate Functions

*DDL and DML commands were used to structure, import, validate, and analyze the banking churn dataset.

Financial Metrics & Statistical Overview



The screenshot shows a PostgreSQL query editor interface. At the top, there are tabs for 'tics', 'Dependencies', 'Dependents', and 'Processes'. Below the tabs is a toolbar with icons for file operations, a search filter set to 'No limit', and a play button. The main area is divided into 'Query' and 'Query History' sections. The 'Query' section contains a SQL query: `1 SELECT AVG(creditscore), AVG(balance)` and `2 From churn;`. At the bottom, there are tabs for 'Data Output', 'Messages', and 'Notifications'. The 'Data Output' tab is active, showing a table with two columns: 'avg numeric' and 'avg numeric'. The first row of data shows the values 650.5288000000000000 and 76485.889288000000.

```
1 SELECT AVG(creditscore), AVG(balance)
2 From churn;
```

	avg numeric	avg numeric
1	650.5288000000000000	76485.889288000000

AVG queries

Average Credit Score: 650.5
Average Balance: 76485.8
Median Balance: 97198.5

Metric	Observation
Avg Credit Score	Indicates customer credibility
Avg Balance	Reflects financial engagement
Mean vs Median Balance	Negatively skewed balance distribution

Customer Segmentation & Churn Patterns



Geography

- Regional churn distribution varied across customer locations.

Spain France Germany



Gender

- Customer exit behavior differed across gender groups.

Female Male



Active Members

- Inactive customers displayed significantly higher churn tendencies.

	exited integer	count bigint
1	0	7963
2	1	2037



Business Insights

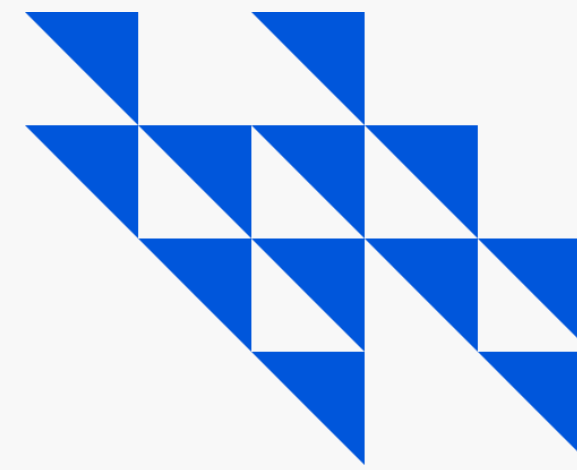
- ❖ Customer inactivity is a major churn indicator.
- ❖ Geography influences retention behavior.
- ❖ Financial engagement impacts loyalty.
- ❖ Early churn identification can reduce revenue loss.



Strategic



Recommendations



Launch targeted engagement campaigns for inactive customers

Implement churn monitoring dashboards using customer activity metrics

Offer personalized financial products for high-value customers

Develop region-focused retention programs





Conclusion

- ❖ This analysis demonstrated how PostgreSQL and SQL-based exploration can transform raw customer data into actionable business insights.
- ❖ The findings highlight the importance of customer engagement, financial behavior, and regional segmentation in reducing churn and improving retention strategies.